Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Signed area

Meaning in my words How would you explain signed area to a friend?	**How can you tell?** What do you do with the area below the time axis when you find displacement?
Example or sketch Sketch a rectangle at −2 m/s lasting 3 s. What displacement does it show?	**Non-example and why** Someone labels the area −2 m/s. Which graph reading did they use?

Distance

Meaning in my words How would you explain distance to a friend?	**How can you tell?** A cart goes 3 m right and 2 m left. How much ground do its wheels cover?
Example or sketch Draw a trip that covers 6 m and finishes at its starting point.	**Non-example and why** A partner writes −5 m for distance. What does the minus sign tell you?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Component

Meaning in my words How would you explain component to a friend?	**How can you tell?** If east and north are positive, which component is negative for a southeast trip?
Example or sketch What two arrows could you use to build a displacement of 3 m east and 4 m north?	**Non-example and why** A classmate counts the diagonal and both parts as three separate trips. What is counted twice?

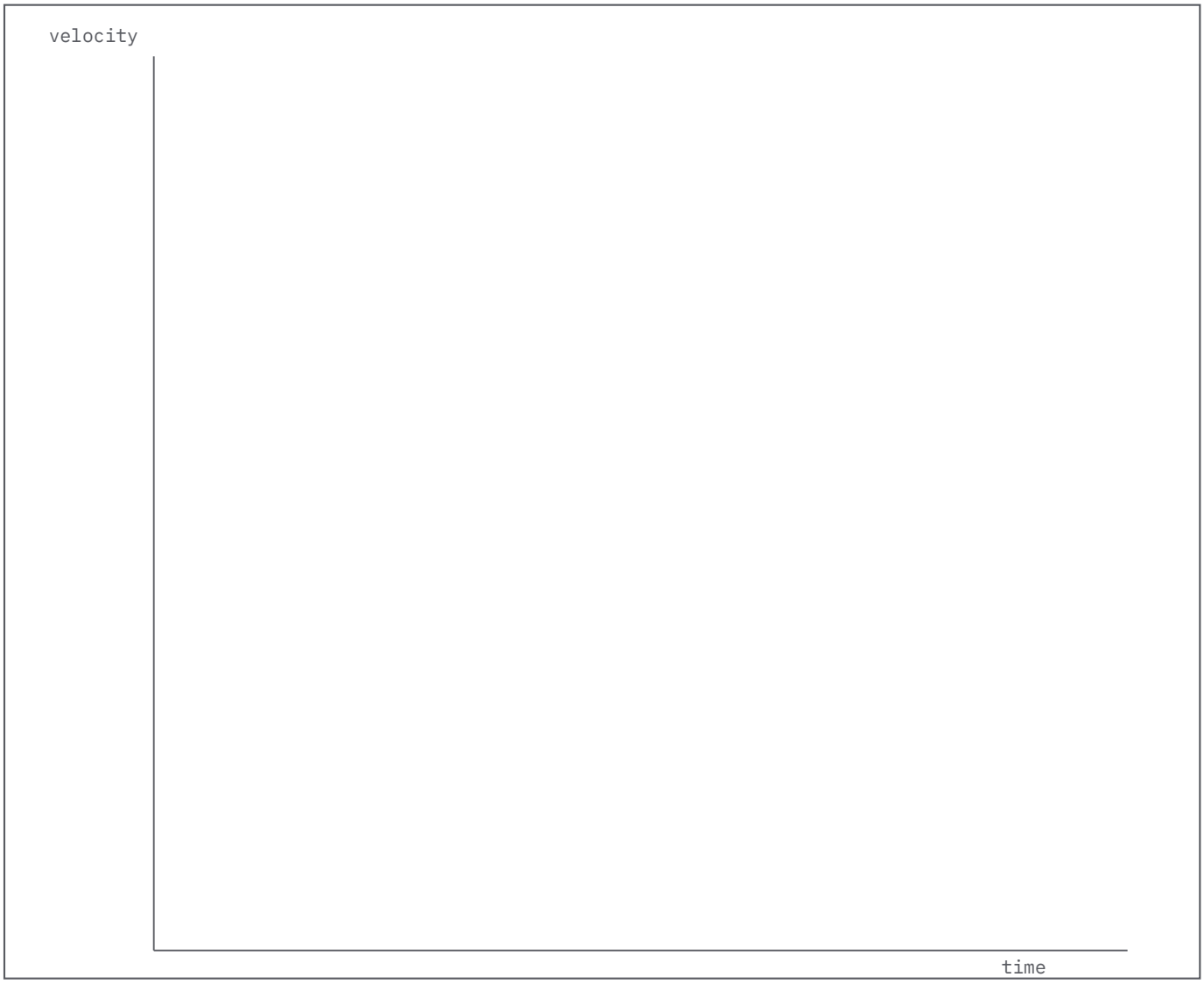
Angle θ

Meaning in my words When you label an arrow with an angle, what does that angle tell someone?	**How can you tell?** What extra information do you need when a vector is labeled “30 degrees”?
Example or sketch Show 30 degrees north of east on a sketch.	**Non-example and why** Does 30 degrees east of north point the same way as 30 degrees north of east?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw and shade opposite-side rectangles. Put m/s on the vertical and s on the horizontal axis.



My marks mean

Where does the camera finish?

04

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Use signed graph areas to find displacement.

CHOOSE ONE JOB TOGETHER

TRACKING ENGINEER ☐

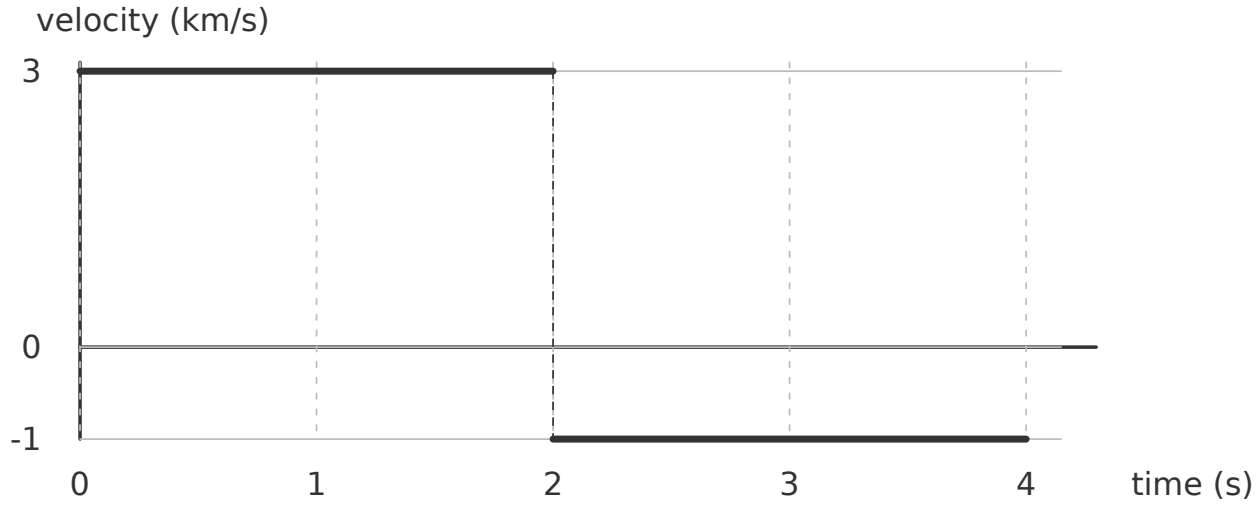
Shade both parts of the velocity graph. Radio the camera's final change in position, including direction.

REPLAY PRODUCER ☐

Mark the outward and return shots. Report total distance and net movement to the editor.

Point to a clue. Keep your first marks; add the update using the color key.

A moving camera follows a model at **+3 km/s** for **2 s**, then **−1 km/s** for **2 s**. The editor needs the end location and the total footage distance.



01 OUR FIRST CALL

Where does the camera finish relative to its start? How far did it travel?

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Check the new caption. Which number belongs on the location graphic? Explain.

Switch speakers. If the camera also moves sideways, why does that motion need its own velocity-time graph?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks